ABT RECERTIFICATION EXAMINATION #18

May 1, 2001 *PART A*

This examination constitutes part of the recertification process for toxicologists certified by the American Board of Toxicology in 1992 and 1997. There are three parts in this exam and an answer sheet is provided for each part. These answer sheets must be returned to ABT, P.O. Box 30054, Raleigh, NC 27622 for grading. An analysis of your performance in each part of the exam will be provided to the recertification committee of ABT and they will contact you regarding this and other requirements of the recertification process. If you have any questions please call the ABT Office at 919-847-8720.

EXAMINATION CATEGORIES

1. Metals, solvents, pesticides, gases, dusts and aerosols, industrial chemicals and radiation, plant and animal toxins, food additives, drugs and cosmetics.
2. Mutagenesis, carcinogenesis, developmental and reproductive toxicology, pulmonary toxicology, neurologic toxicology, immunologic toxicology, cutaneous toxicology, ocular toxicology, hepatic toxicology, hematopoietic toxicology, renal toxicology.
3. General principles, toxicokinetics, toxicodynamics and mechanisms of action, risk assessment, epidemiology and biostatistics, regulatory toxicology and risk management, environmental toxicology, occupational toxicology, forensic and clinical toxicology.

INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet ONLY on line that says name.
5. Do not bend answer sheets.

Good luck!

Key for the following questions
Choose the single most appropriate answer

1. The preferred treatment for mercuric chloride poisoning is
   A) dimercaprol (BAL)
   B) atropine plus pralidoxime (2-PAM)
   C) styramine
   D) calcium disodium edetate (EDTA)
   E) deferoxamine

2. Which of the following glycols is used in pharmaceutical preparations because of its low oral toxicity?
   A) diethylene glycol
   B) trimethylene glycol
   C) propylene glycol
   D) ethylene glycol
   E) none of the above

3. The major life-threatening effect following poisoning with phenol is
   A) optic neuritis with permanent blindness
   B) severe gastrointestinal irritation
   C) esophageal stricture
   D) fatty degeneration of the liver
   E) respiratory depression

4. A farmer arrives at the hospital with the following symptoms: coughing, choking, vomiting, CNS depression and cyanosis. A few hours earlier he was fixing his tractor which had stalled in a field during spring plowing. Clinical findings include albuminuria and hematuria. Which of the solvents would the physician suspect of causing the observed signs?
   A) methanol inhalation from tractor radiator
   B) dermal exposure to carbon tetrachloride used for cleaning
   C) ethanol ingestion
   D) gasoline aspiration from siphoning
   E) dermal exposure to ethylene glycol antifreeze

5. Tricothecene toxins such as T-2 are produced by a species of the
   A) fungi *Fusarium*
   B) bacteria *Clostridium*
   C) virus *Tricothurmus*
   D) dinoflagellate *Tricornus*
   E) algae *Laurencia*

6. The most common enzyme found in snake venoms is
   A) lactate dehydrogenase
   B) NAD nucleosidase
   C) acetylcholinesterase
   D) lecithinase
   E) alanine aminotransferase

7. Which of the following is most likely to cause death once clinical signs are evident despite initiation of treatment?
   A) toxin produced by *Clostridium botulinum*
   B) toxin produced by *Clostridium tetani*
   C) venom produced by the honeybee
   D) venom produced by the brown recluse spider
   E) all are equally likely to produce fatalities

8. Which of the following is a mechanism of toxicity resulting from oral exposure to oxalate-containing plants?
   A) hepatic failure due to metabolites resulting from lung P450 metabolism
   B) destruction of the myelin sheath due to interference with sphingomyelin production
   C) renal failure due to oxalate binding with calcium to produce insoluble crystals
   D) pulmonary edema due to epithelial damage by free radicals
   E) cardiac failure due to blockage of calcium channels

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   D) dinoflagellate *Tricornus*
   E) algae *Laurencia*

10. Which of the following metals is/are nephrotoxic?
   A) mercury
   B) chromium
   C) uranium
   D) B and C
   E) A, B and C

11. Open-range livestock grazing near a fertilizer plant and an agricultural area were diagnosed with blind staggers. They most likely consumed forage
   A) containing a high concentration of fluoride released from phosphate processing
   B) recently sprayed with certain organophosphate insecticides
   C) with mycotoxin contamination due to unusually wet conditions
   D) containing a high concentration of selenium
   E) growing on soil with an imbalance of essential divalent cations (Ca, Mg, Fe)

12. Which one of the following is NOT true of mercury poisoning?
A) ingestion of the inorganic forms is less toxic than the organic forms
B) is most severe when the central nervous system is involved
C) is effectively treated with calcium disodium EDTA
D) the methylated form has been implicated as a human teratogen
E) fish may contain enough mercury to affect humans under some conditions

13. Which one of the following agents produces atrophy of the testicular germinal epithelium?
A) hexane
B) aniline
C) methylethyl ketone
D) toluene
E) carbon disulfide

14. Which of the following forms of lead presents the greatest hazard from dermal exposure?
A) lead sulfide
B) white lead
C) chrome yellow
D) tetraethyl lead
E) black lead

15. Which of the following enhances excretion of zinc and interferes with thiamine utilization?
A) carbon monoxide
B) propane
C) kerosene
D) carbon tetrachloride
E) carbon disulfide

16. Acute copper intoxication
A) interferes with the uptake of zinc and cobalt
B) has been associated with hemolytic anemia in sheep and occasionally humans
C) occurs less frequently in fish and ruminants than in monogastric mammals
D) none of the above
E) all of the above

17. In experimental animals the toxic effect of one metal can be prevented or reversed by the administration of a second metal which itself can induce an adverse effect. In which of the following pairs can the administration of the first metal "protect" against effects of the second?
A) selenium/methyl mercury
B) arsenic/selenium
C) iron/lead
D) zinc/cadmium
E) all of the above

18. The acute toxic effect of C5 to C8 hydrocarbons is
A) renal failure
B) CNS depression
C) liver failure
D) aspiration and hydrocarbon pneumonitis
E) hemolytic anemia

19. After an accident, a worker was determined to have between 400 and 500 cGy of low LET radiation. The worker
A) will die within 48 hours of CNS syndrome
B) will die in less than one week from GI tract syndrome
C) may die from hematopoietic syndrome in 5-8 weeks without intensive medical support
D) will suffer from nausea and vomiting for severalweeks but then should recover
E) will experience no significant symptoms from the exposure

20. Toxic effects of the fumigants dibromochloropropane (DBCP) and ethylene dibromide include
A) gastric squamous cell carcinoma in rodents
B) hemolytic anemia
C) nephrotoxicity
D) delayed neurotoxicity
E) C and D

21. Which of the following rodenticides acts through digitalis-like effects on the heart?
A) sodium monofluoroacetate
B) strychnine
C) arsenic
D) thallium
E) red squill

22. Following ingestion of a rodenticide, a dog exhibited severe tetanic convulsive seizures and died 1 hour later. Postmortem examination revealed cyanosis, dark blood, heart in diastole and subepicardial hemorrhages. The most likely cause of the symptoms and death of this animal is
A) sodium fluoroacetate
B) alpha-naphthylthiourea
C) warfarin
D) diisopropylfluorophosphate
E) strychnine

23. In poisoning by organophosphates, treatment with atropine will
A) release acetylcholine
B) reactivate cholinesterase
C) relieve the noradrenergic toxicity
D) reverse the neuromuscular paralysis
E) relieve the muscarinic symptoms

24. Which of the following pesticides is goitrogenic in laboratory animals?
A) alpha-naphthylthiourea
B) ethylene dibromide
C) dinitroorthocresol
D) phosphine
E) 2,4-dichlorophenoxyacetic acid

25. Of the following five asbestos fiber types, which is the most potent inducer of mesothelioma?
A) crocidolite
B) amosite
C) chrysotile
D) anthophyllite
E) actinolite

26. Two situations which theoretically provide equal hemoglobin capacity for oxygen transport are a) 50% blood carboxyhemoglobin concentration, and b) removal of 50% hemoglobin from the blood. Which of the following comparisons is true?
A) the oxygen content of the blood leaving the lungs is lower in (a) than (b)
B) the oxygen content of the peripheral venous blood is lower in (a) than (b)
C) the oxygen content of the blood leaving the lungs is higher in (a) than (b)
D) the tissue hypoxia is more severe in (a) than (b)
E) the individual oxyhemoglobin dissociation constants are the same in (a) and (b)

27. The initial interaction between ionizing radiation and a cell component that results in cell damage or death is between the radiation and
A) H_2O, resulting in the generation of free radicals
B) RNA, resulting in the disruption of protein synthesis
C) DNA, resulting in the formation of thymine dimers
D) the cell membrane, resulting in changes in ion fluxes
E) polysomes, resulting in inhibition of protein synthesis

28. The primary neoplastic change observed in radium dial painters occupationally exposed to radium via ingestion was
A) leukemia
B) colon carcinoma
C) osteosarcoma
D) neuroblastoma
E) hepatocellular carcinoma

29. Which of the following statements relating to succinylcholine is correct?
A) it is a local anesthetic
B) it is hydrolyzed by plasma cytochrome oxidase
C) it may cause malignant hyperthermia
D) it may cause fatal apnea in patients with atypical plasma cholinesterase
E) C and D

30. Common side effects of spironolactone therapy in man are painful gynecomastia, decreased libido and impotence, which are believed to be due to
A) direct damage to the Leydig cells
B) in part an inhibition of androgen binding
C) stimulation of testicular cytochrome P450
D) chelation of zinc-containing proteins
E) inhibition of binding of aldosterone to the mineralocorticoid receptor

31. A couple ate dinner at a South Florida seafood restaurant. They both had the special which was barracuda. Two hours later they developed symptoms of tingling on their faces and hands, salivation, bradycardia, abdominal pain, nausea, vomiting, watery diarrhea and parasthesias around the face and lips. The most likely cause of these effects was
A) organophosphate pesticide exposure
B) ingestion of ciguatoxin
C) ingestion of botulinum toxin
D) *Staphylococcus* food poisoning
E) scombroid poisoning

32. Intoxication with tricyclic antidepressants causes all of the following except?
A) oxaluria
B) dry mouth
C) anticholinergic effects
D) cardiac arrhythmia
E) urinary retention

33. Examples of direct food additives include all EXCEPT which of the following groups of chemicals?
A) pesticides
B) preservatives
C) flavoring agents
D) nutritional supplements
E) coloring agents

34. Prolonged use of phenytoin in an adult may lead to
A) hypocalcemia
B) osteomalacia
C) elevated plasma alkaline phosphatase activity
D) increased metabolism of vitamin K
E) all of the above

35. The fever which results from excessive absorption of the herbicide dinitroorthocresol is caused by
A) uncoupling of cellular oxidative phosphorylation
B) neurogenic suppression of sweating
C) cutaneous vasoconstriction
D) disturbance of hypothalamic thermoregulatory centers
E) hyperthyroidism

36. Which of the following foods or food additives induce(s) hepatic xenobiotic metabolizing enzymes?
A) butylated hydroxytoluene
B) charcoal-broiled beef
C) brussel sprouts
D) garlic
E) all of the above

37. Disulfiram acts by
A) decreasing the blood level of ethanol
B) decreasing the blood level of acetaldehyde
C) increasing the blood level of ethanol
D) increasing the blood level of acetaldehyde
E) inhibiting alcohol dehydrogenase

Key for the following questions
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

38. Which of the following occupational lung diseases result(s) from an allergic response?
1) byssinosis
2) farmer's lung
3) bagassosis
4) asbestosis

39. An aerosol having a mass median aerodynamic diameter of 4.25 (geometric standard deviation of 1.1) is said to be
1) hygroscopic
2) an aerosol with a high respirable fraction
3) electrophilic
4) nearly monodispersed

40. Regarding the release of oxygen from hemoglobin
1) the oxygen dissociation curve is sigmoidal in shape
2) upon passage of blood through the capillaries, the average tissue normally extracts about 1/4 of the oxygen from hemoglobin
3) carbon monoxide increases the apparent affinity of hemoglobin for oxygen
4) the release of the first oxygen molecule facilitates the release of the second oxygen molecule

41. Silica
1) is toxic to macrophages
2) is considered to be an inert dust
3) can be crystalline or amorphous
4) produces predominantly diffuse alveolar fibrosis

42. Carbon tetrachloride exposure can cause
1) fatty liver
2) myocardial sensitization to catecholamines
3) hepatocellular carcinoma in rodents
4) pulmonary fibrosis

43. Which of the following statements is/are correct?
1) the principal DDT metabolite excreted in man is DDA
2) the principal DDT metabolite stored in body fat is DDE
3) DDE and DDA are less toxic than DDT
4) blood DDE/DDT ratios greater than 1.0 indicate recent DDT exposure

44. Insoluble particles such as $^{239}PuO_2$ and carbon black will accumulate in
1) Type II alveolar cells
2) lymph nodes
3) goblet cells
4) lung macrophages

45. Tamoxifen is
1) an antiandrogen approved for use as a fungicide
2) a mixed agonist/antagonist for estrogenic activity
3) reported to decrease endometrial cancer in women
4) a drug with antiestrogenic properties used for women at risk for breast cancer

46. Terfenadine is one of the drugs associated with QT prolongation in humans. Which of the following statements is/are correct in relation to this syndrome?
1) the QT interval of an ECG trace is a measure of ventricular depolarization
2) prolongation of QT interval can result in lethal ventricular tachycardias, such as *torsades de pointes*
3) the majority of drugs associated with QT prolongation have cardiovascular indications
4) terfenadine is particularly a risk factor if co-administered with drugs which inhibit hepatic CYP3A4

47. Which of the following often become nonintentional food additives during packaging and storage of food?
1) insect and other animal filth
2) parasitic organisms
3) solvents and other chemicals from packaging materials
4) preservatives

48. Clinical illness is commonly associated with overdose of which of the following?
1) vitamin A
2) riboflavin
3) vitamin D
4) thiamine

49. Which of the following statements is/are true?
1) all GRAS substances undergo rigorous laboratory tests prior to approval
2) ingredients that do not meet GRAS criteria cannot be used in food
3) natural products are automatically given GRAS status
4) substances on the GRAS list are subject to review and removal from the list

50. Which of the following commonly used nonprescription drugs may cause liver toxicity in adult humans?
1) dextromethorphan
2) aspirin
3) diphenhydramine
4) acetaminophen

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1. Transgenic animals
   A) have bacteria incorporated into their organs
   B) contain extra DNA sequences in their somatic cells
   C) are used to detect antibodies to genetic material
   D) turn blue in a positive assay
   E) must be fed lactose to survive

2. In a traditional subchronic toxicologic evaluation, what would be the primary methodology to ascertain the presence of an immunotoxic response
   A) bone marrow progenitor cell assays
   B) histopathology and hematology
   C) host resistance assays
   D) lymphoproliferative responses
   E) qualitative immunoglobulin assays

3. d-Limonene, a chemical widespread in natural food stuffs, causes renal tumors in male rats. However, this chemical is NOT considered to pose a cancer risk to humans because the tumors are
   A) histologically different from human renal tumors
   B) benign
   C) highly variable in size
   D) associated with intratubular accumulation of $alpha_2$-microglobin
   E) not found in mice

4. Occupational exposure to benzene may lead to which cancer?
   A) hepatoma
   B) renal carcinoma
   C) leukemia
   D) Hodgkins disease
   E) sarcoma

5. In humans, methemoglobinemia is NOT caused by which of the following materials?
   A) aniline
   B) benzidine
   C) benzene
   D) sodium nitrite
   E) aromatic nitro compounds

6. The life span of peripheral erythrocytes is longest in which of the following species?
   A) mouse
   B) rat
   C) hamster
   D) rabbit
   E) human

7. 2,3,7,8-TCDD exerts immunosuppressive effects on cell-mediated immunity by which of the following proposed mechanisms?
   A) a direct effect on B-cells
   B) a direct effect on T-cells
   C) a direct effect on thymic epithelium
   D) bone marrow depression
   E) all of the above

8. Which of the following statements about exposure to di-(*n*-octyltin dichloride is NOT true?
   A) causes thymic atrophy in rodents
   B) decreases response of B lymphocytes to a mitogenic stimulus
   C) decreases in CD4 positive cells in the peripheral blood
   D) produces no signs of macrophage activation
   E) reduces cell mediated immune responses in the rat

9. Which of the following is true about DNA or protein adducts?
   A) serve as biologic dosimeters in providing estimates of exposure
   B) can provide information on differences between species on rates and routes of metabolic activation and detoxification
   C) ^{32}P-postlabeling is useful for monitoring hemoglobin adducts
   D) A and B
   E) A and C

10. Which of the following is an epithelial neoplasm with cytologic characteristics of malignancy that shows no evidence of invasion?
   A) adenoma
   B) sarcoma
   C) carcinoma *in situ*
   D) pseudocarcinoma
   E) adenocarcinoma

11. Which of the following tumors does NOT occur in high incidence in laboratory animals?
   A) mammary tumors in female Sprague-Dawley rats
   B) testicular tumors in male Fischer 344 rats
   C) leukemia/lymphoma in B6C3F1 mice
   D) liver tumors in male B6C3F1 mice
   E) lung tumors in Sprague-Dawley rats

12. Methylcarbonium ions from a variety of sources, including methyl methanesulfate and dimethylnitrosamine, are thought to elicit mutagenic and carcinogenic effects via which of the following mechanism(s)?
- A) nonspecific, irreversible covalent binding to RNA
- B) initiation of lipid peroxidation
- C) alkylation of guanine bases in DNA at the N-7 position
- D) alkylation of guanine bases in DNA at the O-6 position
- E) A and D

13. His C3076 and his G46 are the primary genotypes of histidine auxotrophic strains of *Salmonella typhimurium*. The basis for such designation is
- A) alphabetical designation in a sequence of histidine mutants isolated
- B) C stands for cysteine and G stands for glutamate modified through a base-pair alteration or frame-shift mutation
- C) C stands for cytidine and G stands for guanidine modified through a base-pair alteration or frame-shift mutation
- D) C and G stand for specific enzymes blocked in histidine synthesis to bring about a mutation
- E) C and G stand for specific genes in histidine operon modified through a base-pair alteration or frame-shift mutation

14. Cyclosporine-induced nephrotoxicity is due to
- A) inhibition of ATP synthesis by a hepatic microsomal metabolite of cyclosporine
- B) drug-induced decrease in glomerular filtration rate
- C) disruption of renal cell calcium homeostasis by a phase II metabolite of the drug
- D) a drug-induced desquamation of tubular cells resulting in a blocking of the tubule
- E) inhibition of renal cortical DNA replication resulting in necrosis

15. Excision repair involves which of the following?
- A) sealing of single strand breaks by DNA glycosylase
- B) an endonuclease which nicks the DNA backbone in the vicinity of the damage
- C) repair by a methyltransferase
- D) the specific cleavage of pyrimidine dimers
- E) none of the above

16. A cellular characteristic of the kidney responsible for the susceptibility to cadmium toxicity is that
- A) renal tubular cells are metabolically very active
- B) proximal tubular cells actively secrete toxicants into tubular urine
- C) proximal tubular cells resorb toxins from urine into blood
- D) renal tubular cells absorb cadmium-metallothionein complex
- E) renal tubular cells require a large oxygen supply

17. Which of the following items is NOT characteristic of light hydrocarbon nephropathy?
- A) hyaline droplet formation in the renal proximal convoluted tubule cells
- B) granular cast formation due to the sloughing of cellular debris into the renal tubular lumen
- C) primarily observed following exposure to saturated aliphatic compounds that contain a ring in their structure
- D) appears to be species and sex dependent
- E) postulated to involve a chemical interaction with alpha$_2$-microglobulin

18. The most probable developmental toxicity resulting from exposure to toxicants during the first week after fertilization is
- A) death of the embryo
- B) growth retardation of the fetus
- C) impaired development of the central nervous system
- D) malformations of the limbs
- E) induction of cancer

19. Which of the following diseases or conditions of the lung is NOT properly paired with its cause?
- A) farmer's lung caused by spores of thermophilic actinomyces
- B) bagassosis caused by dust from dried sugar cane
- C) byssinosis caused by cotton flax or hemp dust
- D) asthma-like symptoms caused by toluene diisocyanate
- E) siderotic lung disease (welder's lung) caused by exposure to acetylene gas

20. Thalidomide ingestion by pregnant women resulted in which effect(s) in their children?
- A) congenital heart disease
- B) phocomelia
- C) renal anomalies
- D) intestinal anomalies
- E) all of the above

21. The majority of human birth defects is attributed to which of the following?
A) genetic abnormalities
B) drugs of abuse
C) environmental pollution
D) unknown etiology
E) over-the-counter medicines

22. Ingestion of ethanol has been shown to potentiate the hepatotoxicity of the following substances EXCEPT
A) carbon tetrachloride
B) aflatoxin B_1
C) chloroform
D) aminoglycoside antibiotics
E) trichloroethane

23. Alpha-naphthylisothiocyanate is
A) a cardiotoxicant drug
B) an experimental hepatotoxicant
C) a toxic industrial dye
D) a P450 inducer
E) an immunosuppressant

24. The primary site of toxicity associated with chronic exposure to acrylamide is the
A) central nervous system
B) peripheral nervous system
C) pancreas
D) liver
E) none of the above

25. Inhaled cadmium chloride ($CdCl_2$) may gain access to the central nervous system, bypassing the normal protective mechanism of the blood-brain barrier, by which of the following?
A) metabolic transformation to CdO
B) damaging the blood-brain barrier
C) binding to carrier proteins
D) uptake by the olfactory neuroepithelium and axonal transport
E) Cd^{2+} ions have no access to the central nervous system

26. Generalized malnutrition or caloric restriction during pregnancy in rats will NOT lead to which of the following outcomes?
A) irreversible delays in CNS maturation
B) thyroid deficiencies
C) gross structural malformations
D) delayed ossification
E) immunodeficiency

27. Eschar formation may be produced by a severe skin irritant. Eschar formation means
A) formation of a crust of necrotic tissue
B) redness of the skin produced by congestion of capillaries
C) a smooth, slightly elevated area on the skin surface which is redder or paler than the surrounding skin
D) a visible collection of pus within or beneath the epidermis
E) production of vesicles having bloody contents

28. For most drugs or chemical toxicants applied to the skin, the one factor that contributes the most to the degree of systemic bioavailability is
A) biotransformation within the skin
B) absorption through the epidermis (stratum corneum)
C) absorption through the dermis (papillary and reticular layers)
D) biotransformation by the liver
E) absorption through the capillary epithelium

29. Given that tetrodotoxin blocks Na^+ channels and cyclodiene insecticides block GABA (gamma-aminobutyric acid) receptors, then
A) both cause convulsions
B) both cause paralysis
C) tetrodotoxin causes paralysis and cyclodienes cause excitation
D) tetrodotoxin causes excitation and cyclodienes cause paralysis
E) both cause smooth muscle relaxation

30. Which of the following statements is NOT true regarding allergic contact dermatitis?
A) haptens must link with epidermal carrier proteins on Langerhans cells to begin the sensitization process
B) metabolic transformation or photoactivation may convert some nonallergens into reactive haptens
C) the intensity of the reaction is directly proportional to the dose applied
D) keratinocytes play a role in amplifying the response of induction and elicitation phases of an allergic response
E) clinically, an allergic response may be indistinguishable from contact irritant dermatitis

31. The cytochrome P450 system of the liver
A) is easily saturable and leads to zero-order kinetics for most drugs
B) can metabolize all drugs and toxins
C) requires 2 electrons per cycle
D) can function with cadmium instead of iron if necessary
E) none of the above

32. The cell type that is primarily responsible for production of collagen in hepatic fibrosis is the
A) Kupffer cell
B) hepatocyte
C) perisinusoidal stellate cell (Ito cell)
D) monocyte, following its differentiation into a macrophage
E) sinusoidal endothelial cell

33. All of the following are associated with the development of cataracts EXCEPT
A) 2,4-dinitrophenol
B) busulfan
C) indomethacin
D) steroids
E) naphthalene

34. The organogenesis period is characterized by division, migration and association of cells into primitive organ rudiments. This period in humans ranges from
A) 6-27 days of gestation
B) 17-38 days of gestation
C) 45-90 days of gestation
D) 21-56 days of gestation
E) 57-180 days of gestation

Key for the following questions
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

35. Which of the following is/are correct?
1) in photoallergic reactions, UV or visible light absorption results in conversion to a product that is a more potent allergen than the parent compound
2) phototoxicity does not require an adaptive immune response
3) photodermatitis caused by parsley, dill or celery is an example of phototoxicity
4) hyperpigmentation of the skin and opacification of the cornea may occur after a photoallergic reaction

36. Which of the following is/are chloracne-producing chemicals?
1) 1,2,4,5-tetrachlorobenzene
2) 2,3,7,8-tetrachlorodibenzo-*p*-dioxin
3) dinitroorthocresol
4) polychlorobiphenyls

37. Some of the aspects of the ICH guidelines on detection of toxicity to reproduction for medicinal products are that
1) studies conducted according to the guidelines are recognized worldwide
2) behavioral assessment of offspring should be performed in both the rat embryo-fetal development and pre- and postnatal studies
3) it is possible to combine the different rat study types if the combination does not affect the interpretation of reproductive endpoints
4) full sperm analysis and staging of seminiferous tubules are required

38. Olfactory epithelium in the nasal passages
1) contains nerve endings associated with the trigeminal nerve
2) has been shown to contain various P450 isozymes
3) if damaged sufficiently, may not function in olfaction
4) does not absorb water-soluble gases efficiently

39. 1-Methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP) is neurotoxic to dopaminergic neurons in the central nervous system. Which of the following is/are correct?
1) MPTP produces symptoms characteristic of Parkinson's disease in humans, monkeys and pigs
2) MPTP is selectively neurotoxic to dopaminergic neurons in the cerebellum
3) MPTP is activated to the neurotoxic metabolite 1-methyl-4-phenylpyridinium ion by monoamine oxidase
4) 1-methyl-4-pyridinium ion is neurotoxic when administered to animals

40. Neuropathies from the solvent hexane are
1) the same as those caused by 2-hexanone and 2,5-hexanedione
2) distal axonopathies
3) characterized by motor weakness in the extremities, due in part to axonal swelling and demyelination
4) produced by acute exposures

41. Which of the following species show symptoms of tri-*ortho*-cresylphosphate (TOCP) neuropathy closely similar to those in humans?
1) monkey
2) chicken
3) dog
4) cat

42. Which of the following is/are true regarding pulmonary volumes and pulmonary disease?
1) lung volumes in order of increasing volume are: tidal volume, vital capacity, total lung capacity
2) fibrosis is a restrictive lung disease characterized by deposition of excess collagen
3) a plethysmograph is used to measure total lung volume and functional residual capacity and may also be used to measure respiratory frequency and tidal volume
4) in obstructive lung disease such as emphysema, lung compliance and total alveolar surface area are increased

43. Which of the following is/are correct concerning chemical exposures of the eye?
1) the one best treatment for acid burns to the eye is rapid irrigation with large volumes of water
2) the amount of ocular damage produced by an acid depends entirely on the pH of the solution
3) although ocular burns associated with alkalies may appear mild at the time of injury, they may have serious late effects such as opacification, vascularization, ulceration or perforation of the cornea
4) of all the cations measured, sodium ion as sodium hydroxide penetrates epithelium, stroma and anterior chamber more rapidly than any other

44. A reasonable battery of serum enzyme tests to assess hepatic injury in laboratory animals could include
1) aminotransferase
2) lactate dehydrogenase
3) sorbitol dehydrogenase
4) acid phosphatase

45. Which of the following statements is/are correct regarding nephrotoxicants?
1) most toxic agents affect the glomerulus
2) the unusual susceptibility of the kidney can be related to the high renal blood flow delivered to this organ
3) injury by toxins is rarely produced in the proximal tubule
4) gentamicin and cephalosporins are drugs producing nephrotoxicity

46. The kidney is capable of maintaining function even though subjected to repeated non-fatal chemical insults. The basis(es) of this protection against toxic insults include
1) compensatory renal hypertrophy
2) induction of heat shock proteins
3) induction of metallothionein
4) genesis of new nephrons

47. Morphological signs of nephrotoxicity are often localized to the cells of the proximal tubule. This may reflect
1) specific uptake and binding of toxic agents to apical surfaces of certain cell types
2) cell-specific biotransformation capabilities
3) cell-specific transport mechanisms
4) vasopressin-insensitive water reabsorption

48. The pathogenesis of neoplasia is thought to consist of three stages: initiation, promotion and progression. Which of the following characteristics is/are associated with promotion?
1) reversible at both the level of gene expression and at the cellular level
2) dose response exhibits a measurable threshold
3) sensitive to age, dietary and hormonal factors
4) morphologically discernible alterations from karyotype instability

49. Major findings that are considered useful for the evaluation of hepatic injury in laboratory animals include which of the following?
1) elevations of cytochrome P450
2) abnormal hepatic excretory test (e.g., indocyanine green clearance)
3) a significant incidence of binucleated parenchymal cells
4) histopathologic evidence of liver injury

50. Aflatoxin B_1
1) frequently occurs in stored grains
2) is subject to metabolic activation by cytochrome P450
3) acts synergistically with hepatitis to cause cancer
4) is synthesized by corn, peanuts and legumes

ABT RECERTIFICATION EXAMINATION #18

May 1, 2001 *PART C*

This examination constitutes part of the recertification process for toxicologists certified by the American Board of Toxicology in 1992 and 1997. There are three parts in this exam and an answer sheet is provided for each part. These answer sheets must be returned to ABT, P.O. Box 30054, Raleigh, NC 27622 for grading. An analysis of your performance in each part of the exam will be provided to the recertification committee of ABT and they will contact you regarding this and other requirements of the recertification process. If you have any questions please call the ABT Office at 919-847-8720.

EXAMINATION CATEGORIES

1. Metals, solvents, pesticides, gases, dusts and aerosols, industrial chemicals and radiation, plant and animal toxins, food additives, drugs and cosmetics.
2. Mutagenesis, carcinogenesis, developmental and reproductive toxicology, pulmonary toxicology, neurologic toxicology, immunologic toxicology, cutaneous toxicology, ocular toxicology, hepatic toxicology, hematopoietic toxicology, renal toxicology.
3. General principles, toxicokinetics, toxicodynamics and mechanisms of action, risk assessment, epidemiology and biostatistics, regulatory toxicology and risk management, environmental toxicology, occupational toxicology, forensic and clinical toxicology.

INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet ONLY on line that says name.
5. Do not bend answer sheets.

Good luck!

Key for the following questions
Choose the single most appropriate answer

1. If 91% of a weak basic chemical is in the protonated form in the blood, the pKa of that chemical is
   A) 3.8
   B) 6.9
   C) 7.4
   D) 8.4
   E) 9.4

2. Which statement best describes the relationship between air pollution and mortality?
   A) with the exception of industrial disasters, there has never been a clear link between air pollution and mortality
   B) at current levels, there is no link between air pollution and mortality
   C) at current levels, chronic exposure to polluted air may increase mortality but there are no detectable acute effects
   D) at current levels, acute exposure to polluted air causes increased mortality in susceptible populations
   E) at current levels, acute exposure to polluted air causes increased mortality in healthy persons

3. The cofactor used by sulfotransferases is
   A) acetyl coenzyme A
   B) nicotinamide adenine dinucleotide phosphate (NADPH)
   C) phosphoadenosine-phosphosulfate (PAPS)
   D) S-adenosylmethionine
   E) uridine diphosphate glucuronic acid (UDPGA)

4. The selective toxicity of sulfonamide drugs to bacteria is caused by
   A) differences in absorption between humans and bacteria
   B) long half-life of sulfonamides in bacteria
   C) antagonism of the incorporation of *p*-aminobenzoic acid into the folic acid molecule in bacteria
   D) differences in body weight between humans and bacteria
   E) none of the above

5. Which is the principal health hazard in the rayon industry?
   A) carbon monoxide
   B) nitrous oxide
   C) carbon disulfide
   D) sulfur dioxide
   E) toluene

6. The most significant factor with respect to the mobility in soil of PCBs is the
   A) moisture content
   B) average temperature
   C) average humidity
   D) soil organic content
   E) average wind speed

7. Bioaccumulation of chemicals in the food chain is most likely to occur for compounds that display
   A) high lipophilicity
   B) high chemical and metabolic stability
   C) high tendency to bind to humic acids
   D) A and B
   E) A, B and C

8. In ecotoxicology, an ideal sentinel species
   A) is most abundant in the location of interest
   B) is at the top of the food chain
   C) contains a large, consistent amount of biomarkers
   D) is exceptionally sensitive to a pollutant's effects
   E) is opportunistic in its choice of habitats

9. The proper statistical procedures to be used for evaluating a toxicological study are
   A) chosen after completing the study and inspection of the data for differences between treatment groups (post hoc)
   B) not dependent upon the biological significance of a result
   C) chosen after removing outliers from the data set
   D) the same procedures whether continuous or non-parametric data are completed
   E) not dependent on the randomized assignment of animals to treatment groups

10. Which of the following responses in humans is least likely to be predicted from animal studies?
   A) allergic reactions
   B) carcinogenicity
   C) idiosyncratic reactions
   D) developmental toxicity
   E) behavioral effects

11. All of the following environmental contaminants would be expected to accumulate in body fat EXCEPT
   A) 2,2',4,4',5,5'-hexachlorobiphenyl
   B) N-nitrosodimethylamine
   C) DDT
   D) hexachlorophene
   E) chlordane

12. The treatment of acute toxicosis should be carried out
A) only after an accurate diagnosis of the toxin is made
B) by first anesthetizing the patient
C) by first doing a gastric lavage
D) on the basis of the symptomatology
E) after securing blood samples for chemical analysis

13. What percentage of plasma water reaching the kidney is filtered on each pass?
A) 98%
B) 75%
C) 33%
D) 10%
E) 5%

14. Which of the following is/are true regarding drug binding to plasma proteins?
A) acidic drugs bind mostly to albumin
B) acidic drugs bind mostly to alpha-1 acid glycoproteins
C) binding limits glomerular filtration of the drug
D) drug binding is highly selective and independent of the physical-chemical characteristics of the drug
E) A and C

15. Which one of the following is NOT characteristic of both active transport and facilitated diffusion?
A) carrier-mediated transport
B) substrate specificity
C) saturability of the transport system
D) transport at a rate greater than by simple diffusion
E) expenditure of metabolic energy required for transport

16. If a drug has a greater action on the central nervous system of a fetus than that of the mother, one might expect that the drug
A) has a high lipid solubility
B) does not readily penetrate the blood brain barrier
C) is not metabolized by oxidative enzymes of the liver
D) has a low molecular weight
E) is bound to plasma protein of the fetus

17. Cytochrome P4501A1 is most readily induced by
A) phenobarbital
B) ethanol
C) 3-methylcholanthrene
D) DDT
E) clofibrate

18. A compound injected intravenously into rats was found to exhibit a toxicokinetic profile consistent with a single-compartment model and first-order elimination. The apparent volume of distribution (V_d) was much larger than the actual body volume. This finding regarding V_d indicates that the compound
A) actually follows zero-order kinetics
B) is rapidly incorporated into the bone matrix
C) undergoes first-pass hepatic metabolism
D) is rapidly excreted in the urine
E) may have a high affinity for tissue binding sites

19. How many half-lives does it take for a continuously infused xenobiotic to reach 95% of a plateau plasma concentration?
A) 1
B) 2
C) 5
D) 10
E) greater than 10

20. The lower stomach acidity of infants less than 4 months of age permits increased intestinal growth of *E. coli* which can reduce nitrate in foods to nitrite, leading to
A) goiter
B) hirsutism
C) methemoglobinemia
D) hemoglobinemia
E) peptic ulcers

21. Which statement(s) is/are NOT correct?
A) case-fatality rate is the number of deaths from a given cause divided by the number of persons ill with that condition
B) cause-specific mortality rate is the number of deaths attributed to a given cause divided by the population
C) attack rate is the proportion of persons who become ill divided by the number who were at risk or exposed
D) A and C
E) A, B and C

22. Which Federal agency of the United States has the authority to promulgate safety standards for chemicals without the authority to mandate testing to prove safety?
A) Environmental Protection Agency
B) National Institute for Occupational Safety and Health
C) Food and Drug Administration
D) Occupational Safety and Health Administration
E) A and B

23. Which of the following selectively blocks glycine receptors in spinal ganglia?
A) picrotoxin
B) bicuculline
C) tetrodotoxin
D) pyrethrins
E) strychnine

24. Apoptosis (programmed cell death) is characterized by all of the following EXCEPT
A) energy consumption
B) condensation of nuclear chromatin
C) cellular and nuclear swelling
D) internucleosomal cleavage of DNA
E) activation of highly conserved signal transduction pathways

25. Which of the following parameters reflects best the occurrence of a disease at a specific point in time?
A) incidence rate
B) prevalence rate
C) mortality rate
D) age-specific mortality rate
E) case fatality rate

26. In deriving a chronic oral reference dose (RfD) in accordance with EPA guidelines, which of the following extrapolations does NOT usually require an uncertainty factor?
A) LOAEL to NOAEL extrapolation
B) male to female extrapolation
C) subchronic to chronic extrapolation
D) animal to human extrapolation
E) typical population to sensitive subpopulation

27. The hazard index is
A) used to estimate carcinogenic risk
B) the ratio of a therapeutic dose of a chemical to a toxic dose of that compound
C) the ratio of an estimated daily intake of a chemical to a reference dose of that compound
D) a soil clean-up goal at a hazardous waste site
E) equivalent to the therapeutic index for drugs

28. The ratio of the incidence in exposed workers compared to the incidence in unexposed workers is the
A) relative attributable risk
B) relative excess risk
C) relative risk
D) odds ratio
E) none of the above

29. The functions of U.S. regulatory agencies that protect citizens from the hazards of exposure to chemicals include
A) rule making, surveillance and implementation of regulation by the EPA, FDA, NIOSH and DOT
B) NIOSH mandates the testing of chemicals to which there is occupational exposure
C) pesticides, nuclear materials and tobacco (but not foods, drugs or cosmetics) are regulated under EPA's Toxic Substance Control Act (TSCA)
D) EPA under the Federal Insecticide, Fungicide and Rodenticide Act permits registration of a pesticide even if it is a carcinogen provided benefits outweigh risks
E) the National Cancer Institute carries out the bioassay and cancer risk assessment of potential environmental carcinogens under TSCA

30. Encephalopathy occurring during treatment of tuberculosis with isoniazid is attributed to
A) drug-induced sensitization
B) disease-induced cytotoxic hypoxia
C) drug-induced vitamin B_6 deficiency
D) drug-induced cytotoxic hypoxia
E) breakdown of the blood-brain barrier

31. Which of the following occupational/environmental diseases and agents are NOT paired correctly?
A) polymer fume fever - fluorinated polymers
B) pseudo-Parkinsonism syndrome - manganese
C) silo-filler's disease - moldy hay
D) Minamata disease - methyl mercury
E) printer's ink peripheral neuropathy - methyl butyl ketone

32. Hepatic angiosarcoma in humans is most often associated with exposure to which of the following?
A) arsenic
B) coke oven emissions
C) vinyl chloride
D) beryllium
E) talc

33. Which of the following indicates nonlinear pharmacokinetics?
A) the decline in the blood concentration of the chemical follows an exponential function
B) the half-life for elimination of the chemical increases with increasing dose
C) the area under the plasma concentration versus time curve (AUC) is proportional to the dose
D) the amount of the chemical eliminated during a specified time-period increases with increasing dose
E) none of the above

34. Which of the following is NOT true about salicylate poisoning in small children?
A) a child suffering from salicylate poisoning may exhibit hyperthermia
B) oil of wintergreen can cause salicylate poisoning
C) children suffering from severe salicylate poisoning will exhibit respiratory alkalosis
D) children suffering from severe salicylate poisoning will exhibit metabolic acidosis
E) children poisoned by salicylate will have a sweet almond odor to their breath

35. A ten year old, 40 kg child ingested 101 ml of a 40% ethanol (specific gravity - 0.79) solution. If the volume of distribution (Vd) for ethanol is 0.7 l/kg, what is the predicted peak blood ethanol level assuming instantaneous distribution?
A) 1.14 g/l
B) 2.85 g/l
C) 0.114 g/l
D) 1.14 g/ml
E) 2850 mg/ml

36. Which of the following is a transmembrane protein involved in ATP-dependent efflux of various structurally unrelated anticancer drugs which may be responsible for multidrug resistance?
A) a_1 acid glycoprotein
B) P-glycoprotein
C) C reactive protein
D) spectrin
E) glycophorin C

37. Acetaminophen toxicity is due to
A) metabolism to a sulfate conjugate
B) metabolism to a glucuronide conjugate
C) metabolism to a quinoneimine
D) metabolism to a mercapturic acid conjugate
E) acetaminophen itself and not a metabolite

38. Toluene diisocyanate toxicity is normally manifested first in
A) liver dysfunction
B) kidney dysfunction
C) blood dyscrasia
D) bronchoconstriction
E) hypoxemia

39. Which of the following compounds is the principal lipid-soluble antioxidant found in cell membranes?
A) alpha-tocopherol (vitamin E)
B) ascorbic acid (vitamin C)
C) carotenoids
D) tannins
E) glutathione

Key for the following questions
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

40. Which of the following is/are true of the U.S. Toxic Substance Control Act?
1) the EPA may require manufacturers to conduct testing on substances based on production volumes alone
2) TSCA requires the EPA to evaluate a new chemical 90 days before it is placed in commerce
3) the Premanufacture Notification (PMN) includes a minimum data set
4) new substantial risk information must be reported immediately to the Administrator

41. When a chemical is administered repeatedly at a fixed dose and dosing interval, its concentration in an organ at steady state will
1) increase linearly with continued dosing
2) not change with continued dosing
3) increase more slowly than at the beginning of the dosing regimen
4) depend on the rates of dosing and clearance

42. GLP regulations require which of the following?
1) each individual involved in a nonclinical laboratory study shall have appropriate education, training and experience to perform the assigned functions
2) a study director will be assigned for each study
3) a testing facility shall have a quality assurance unit
4) the study director must have a M.S. or Ph.D. degree

43. According to Good Laboratory Practices regulations, pathologists involved in animal toxicology studies should
1) read slides in a "blind" manner
2) have all the in-life data available when reading slides
3) have the slides presented in a randomized manner
4) may or may not have the in-life data available depending on the discretion of the study director

44. Teratology studies
1) are generally required for drugs
2) are required in two species
3) are required in rodent and non-rodent
4) involve sacrifice of the female and delivery of fetuses by hysterectomy

45. For an agent eliminated by metabolism, saturation of biotransformation results in which of the following?
1) zero-order kinetics
2) first-order kinetics
3) elimination of a constant amount per unit time
4) elimination of a constant fraction of the amount remaining per unit time

46. The term ppm is equivalent to
1) parts per million
2) mg of x/1,000,000 mg
3) mg of x/kg
4) 0.0001% of x

47. Emetics are contraindicated in cases in which the patient
1) has ingested a strong acid or alkali
2) has ingested a cleaning fluid
3) is in a coma or is semiconscious
4) has ingested large doses of sedative hypnotics

48. A dose of 50 mg of a drug is given to a 50 kg patient by bolus intravenous injection and disappears by a first-order process. At the end of the injection, the plasma concentration is 0.5 ug/mL and after 12 hours the plasma concentration is 0.062 ug/mL. Therefore, the
1) half-life of the drug is 4 hours
2) volume of distribution of the drug is 50 liters
3) volume of distribution of the drug is 100 liters
4) half-life of the drug is 3 hours

$\frac{dose}{pl\cdot c} = \frac{50}{0.5} = 100$

49. To approve a human therapeutic product produced by biotechnology processes, the U.S. Food and Drug Administration would likely require
1) a toxicity test in a single species
2) extensive testing and validation of the purification of the product including tests for the presence of DNA and several viruses
3) only safety tests in mice and guinea pigs on the batches of finished drug as is routinely done for vaccines
4) some of the same animal toxicity studies as for the drug made by nombiotechnological processes

50. In a two-compartment model the
1) semilogarithmic plot of tissue concentration versus time after exposure yields a nonlinear curve
2) slope of the initial phase of the plasma concentration versus time curve can be resolved into a straight line using the method of residuals
3) plasma concentration of the chemical in the distribution phase decreases more rapidly than in the postdistribution beta-phase
4) rate constants (k_{12} and k_{21}) cannot be calculated from a single-dose study